OMTON, Volume *34*

Supplemental information

Boosting CAR T cell functionality with oncolytic viruses for the treatment of pediatric diffuse midline gliomas

Konstantinos Vazaios, Antonio C. Tallon-Cobos, Louisa P.J. van Oosterhout, Piotr Waranecki, Annelisa M. Cornel, Noël M.M. Dautzenberg, Marliek van Hoesel, Caitlyn Forbes, Stefan Nierkens, Vera Kemp, Rob C. Hoeben, Jasper van der Lugt, Friso G. Calkoen, and Esther Hulleman

Supplemental data

Table S1. Differential expression analysis results

A. **Differentially expressed genes after Goravir infection**
B. **Differentially expressed genes after R124 infection**

Table S2. Reactome pathway analysis results

A. **Reactome pathways using the significantly differentiated genes after Goravir infection**
B. **Reactome pathways using the significantly differentiated genes after R124 infection**

Table S3. Effector to target ratios per cell culture used for the combination comparisons in Figures 3B, 3C & 4.

DMG culture	E:T of B7H3-CAR T-cells	E:T of GD2-CAR T-cells
SU-DIPG-IV	1:10	1:20
VUMC-DIPG-G	1:5	1:5
JHH-DIPG-01	1:2	1:5
HSJD-DIPG-07	1:10	1:10
VUMC-DIPG-10	1:2	1:2
VUMC-DIPG-11	1:5	1:5

Diffuse midline glioma (DMG), Effector to target (E: T), Chimeric antigen receptor (CAR).

Table S4. Cytokines and chemokines measured from supernatants across diffuse midline glioma co-cultures and monocultures.

A. **Concentrations (pg/mL) of cytokines and chemokines detected in supernatants of anti-B7H3 and anti-GD2 CAR T-cell co-cultured with uninfected, Goravir and R124-infected diffuse midline gliomas.**

B. **Concentrations (pg/mL) of cytokines and chemokines detected in supernatants of uninfected, Goravir and R124-infected diffuse midline glioma monocultures.**

Table S5. BioLegend capture beads for chemokine/cytokine analysis of co-culture supernatants.

Capture Bead	Cat #	Dilution
Human IL-6 (A3)	741096	1:14
Human CCL2 (MCP-1) (A4)	741097	1:14
Human IL-1β (A5)	741115	1:14
Human IFN-α2 (A6)	741099	1:14
Human CCL5 (RANTES) (A7)	741109	1:14
Human sCD25 (IL-2Ra) (A8)	741117	1:14
Human IL-4 (A10)	741118	1:14
Human IL-7 (B2)	741102	1:14
Human IL-17A (B3)	741120	1:14
Human CXCL8 (IL-8) (B4)	741104	1:14
Human TNF-α (B5)	741105	1:14
Human CXCL10 (IP-10) (B6)	741106	1:14
Human CCL3 (MIP-1α) (B7)	741107	1:14
Human IL-10 (B9)	741108	1:14

Table S6. Antibody mix for spectral flow cytometry analysis of CAR T-cell populations.

A. extracellular staining

Marker	Fluorophore	Supplier	Cat #	Dilution
CD45-RA	BUV395	BD	740298	1:200
CD3	BUV496	BD	612940	1:50
CCR9	BV421	Biolegend	358913	1:100
CD25	BV605	Biolegend	302632	1:50
CD95	BV711	Biolegend	305644	1:200
PD-1	BV750	Biolegend	329966	1:25
CXCR3	BV785	Biolegend	353738	1:25
CTLA-4	PerCP-eFluor 710	Thermo	46-1529-42	1:20
CCR7	PE/Dazzle594	Biolegend	353236	1:50
CD69	PE/Fire640	Biolegend	310960	1:50
TIM-3	PE/Cy5	Biolegend	345052	1:200
CD45	PE/Fire700	Biolegend	304078	1:100
LAG-3	PE/Cy7	Biolegend	369310	1:400
CCR4	PE/Fire810	Biolegend	359433	1:200
CD127	cFluorR720	Cytek	RC-00009	1:100
CD56	APC/Fire750	Biolegend	392408	1:400
CD8	APC/Fire810	Biolegend	344764	1:50
CCR5	BUV737	BD	748873	1:100
HLA-DR	BV570	Biolegend	307638	1:50
CD49d	PerCP-Cy5.5	Biolegend	304312	1:20
CD29	PE	Biolegend	303004	1:100

CD137	PerCP/Fire 806	Biolegend	309847	1:100
TIGIT	BB700	Biolegend	747846	1:25
B. Intracellular staining				
CD4	BUV805	BD	742000	1:50
GrzmB	Pacific Blue	Biolegend	515408	1:100
ki67	BV650	Biolegend	563757	1:50
IFN-γ	Spark NIR685	Biolegend	502552	1:50

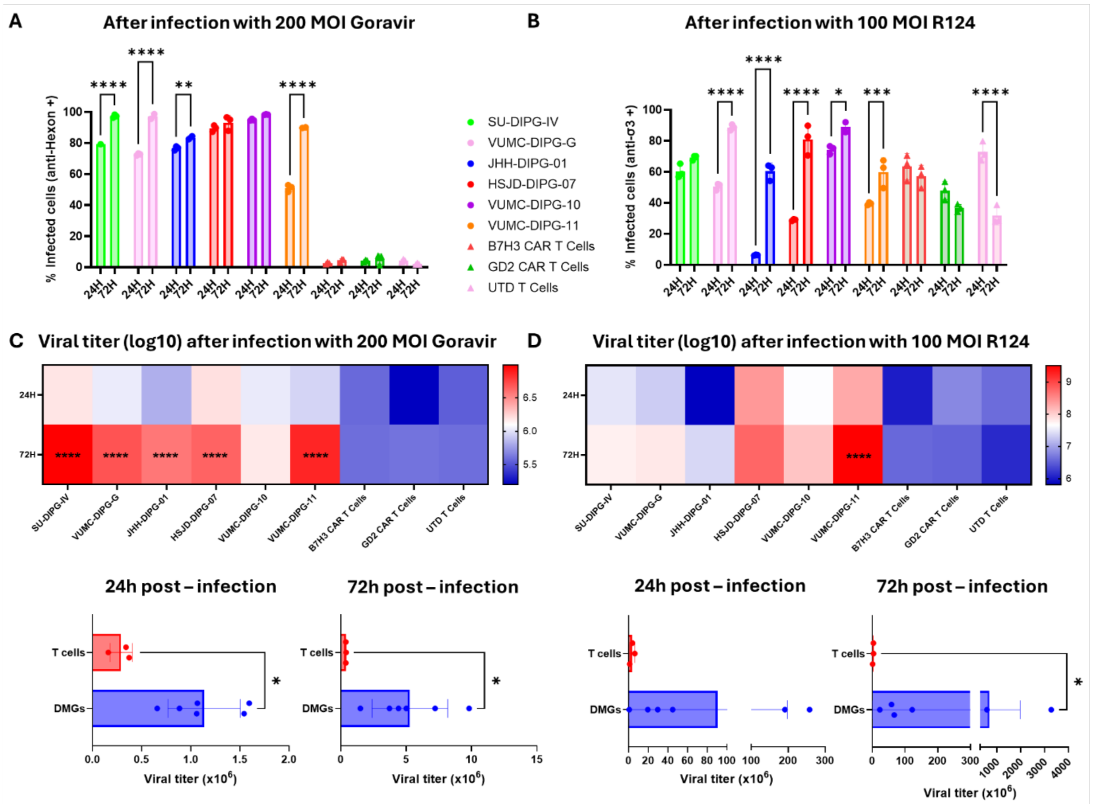


Figure S1. Viral infection and replication are restricted to tumor cells, with limited infection and replication in non-tumor cells even at high MOIs.

Patient-derived diffuse midline glioma (DMG) cultures, CAR T-cells and untransduced T-cells (UTD) were incubated with the indicated multiplicity of infection (MOI) of Goravir or R124 for 24 and 72 hours. Comparisons of **A)** Goravir (hexon+) and **B)** R124 (σ3+) infected cells out of the total alive cell population of each culture after 24 and 72 hours of infection.Viral titer comparison of **C)** Goravir and **D)** R124 per cell culture between 24 and 72hours of infection and across DMG and T-cells (n=3 independent experiments). For **A** and **B,** each bar and error bar represent the mean ± SD of n=3 biological replicates. Statistics were measured with an unpaired t-test per culture. For **C** and **D,** each heatmap box indicates the mean ± SD of log10-transformed viral titer values (n=3 independent experiments), while each bar and error bar represent the mean ± SD of viral titer (x10^6) (n=6 DMG and n=3 T-cell cultures). Statistics were assessed with an unpaired t-test per culture for the heatmaps and with the Mann-Whitney test for the box plots. * , p<0.05; **, p<0.01; ***, p<0.001; ****, p<0.0001.

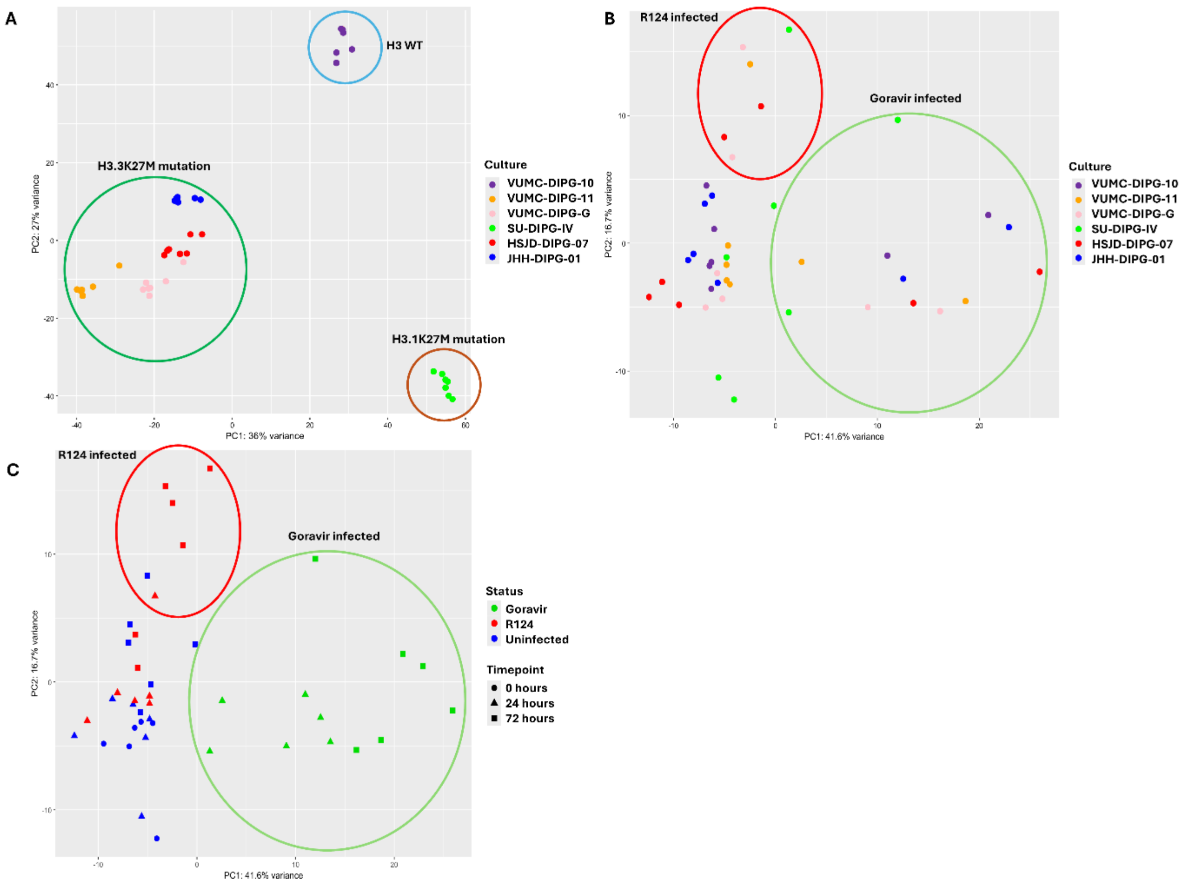


Figure S2. OV-driven changes are masked by the highly heterogeneous expression background of the DMG cultures.

Principal component analysis plot with Principal component (PC) 1 as x-axis and PC2 as y-axis of DMG cultures consisting of samples uninfected (n=18, 3 per DMG culture), infected with Goravir (n=12, 2 per DMG culture) or infected with R124 (n=12, 2 per DMG culture) for 0, 24 and 72 hours. Each dot represents a sample. **A**) Unsupervised clustering using the top 500 most variable genes. H3 wild-type (WT): VUMC-DIPG-10; H3.3K27M mutation: VUMC-DIPG-11, VUMC-DIPG-G, HSJD-DIPG-07 and JHH-DIPG-01; H3.1K27M mutation: SU-DIPG-IV. Each dot is colored by culture identity. **B**) Unsupervised clustering using the top 500 most variable genes after batch removal of the DMG identity effect and colored by culture identity or **C**) Unsupervised clustering using the top 500 most variable genes after batch removal of the DMG identity effect, colored by infection status: Uninfected (blue), Goravir infected (green) and R124 infected (red) and shaped by time of collection: 0 hours (circle), 24 hours (triangle) and 72 hours (square).

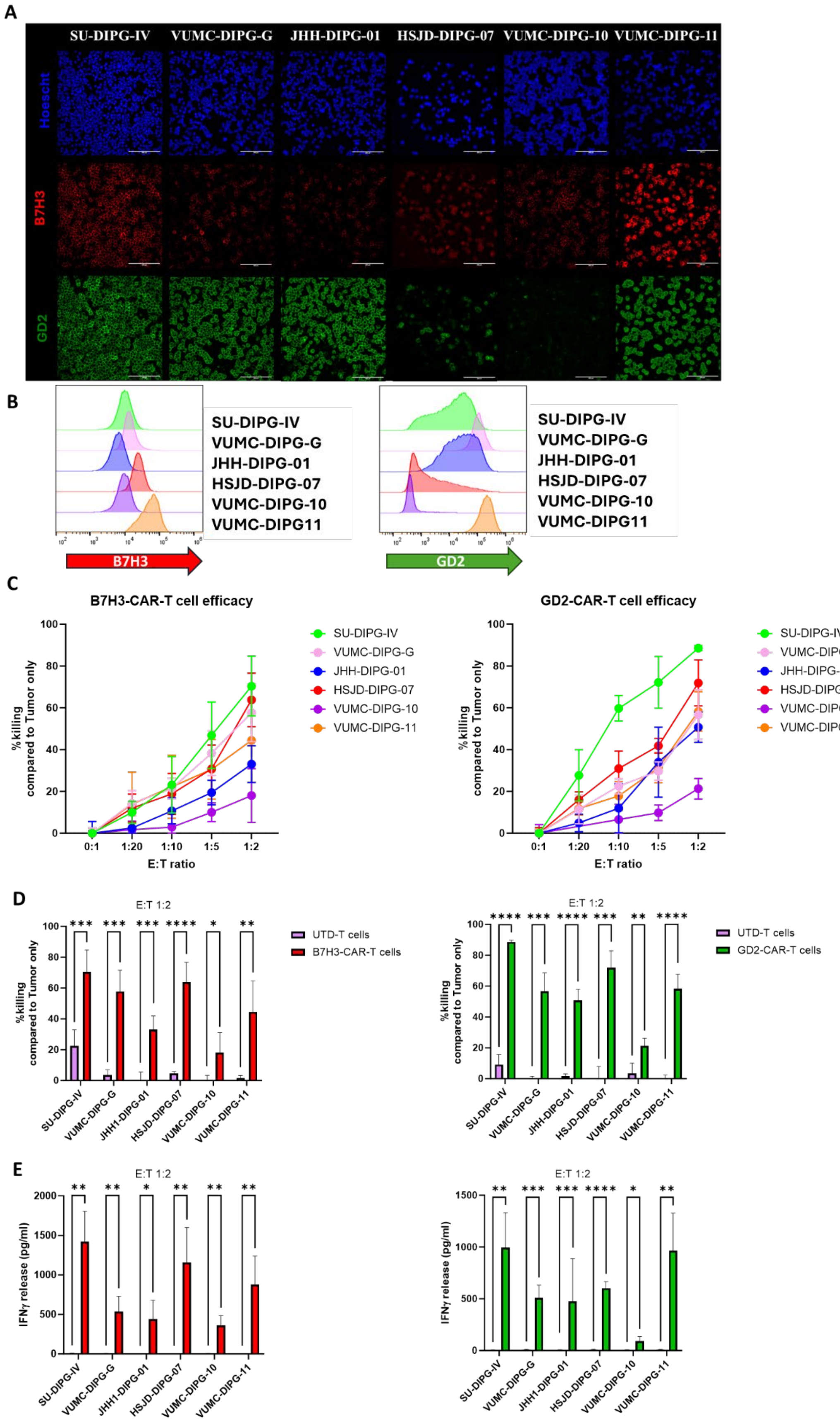

A
SU-DIPG-IV
VUMC-DIPG-G
JHH-DIPG-01
HSJD-DIPG-07
VUMC-DIPG-10
VUMC-DIPG-11
Hoescht
B7H3
GD2
B
SU-DIPG-IV
VUMC-DIPG-G
JHH-DIPG-01
HSJD-DIPG-07
VUMC-DIPG-10
VUMC-DIPG11
B7H3
GD2
C
B7H3-CAR-T cell efficacy
GD2-CAR-T cell efficacy
% killing compared to Tumor only
E:T ratio
0:1
1:20
1:10
1:5
1:2
D
E:T 1:2
UTD-T cells
B7H3-CAR-T cells
GD2-CAR-T cells
%killing compared to Tumor only
E
IFNγ release (pg/ml)

Figure S3. B7H3 and GD2 CAR T cells induce cytotoxicity against DMG cultures.

A) Representative immunofluorescent images of the six DMG cells, scale bar is 200µm. **B**) Representative mean fluorescent intensity (MFI) of B7H3 (top) and GD2 (bottom) of the six DMG cells. **C**) Anti-tumor activity of B7H3 and GD2 CAR T-cells measured by *in vitro* co-cultures of increasing effector target ratio (E: T), with E: T of 0:1 set as 0% CAR T-cell specific killing. Dots and error bars represent the mean ± SD values of n=5 for B7H3 and n=4 for GD2 per E: T and culture. **D)** Anti-tumor activity and **E**) IFN-γ release detected by IFN-γ ELISA as pg/mL of B7H3 CAR T-cells or GD2 CAR-T cells compared to untransduced T-cells (UTD) in E: T of 1:2 per DMG culture co-culture. Bars and error bars represent the mean ± SD values of n=5 for B7H3 and UTD and n=4 for GD2 and UTD per E: T and culture. Statistics were measured by unpaired t-tests. * , p<0.05; **, p<0.01; ***, p<0.001; ****, p<0.0001.

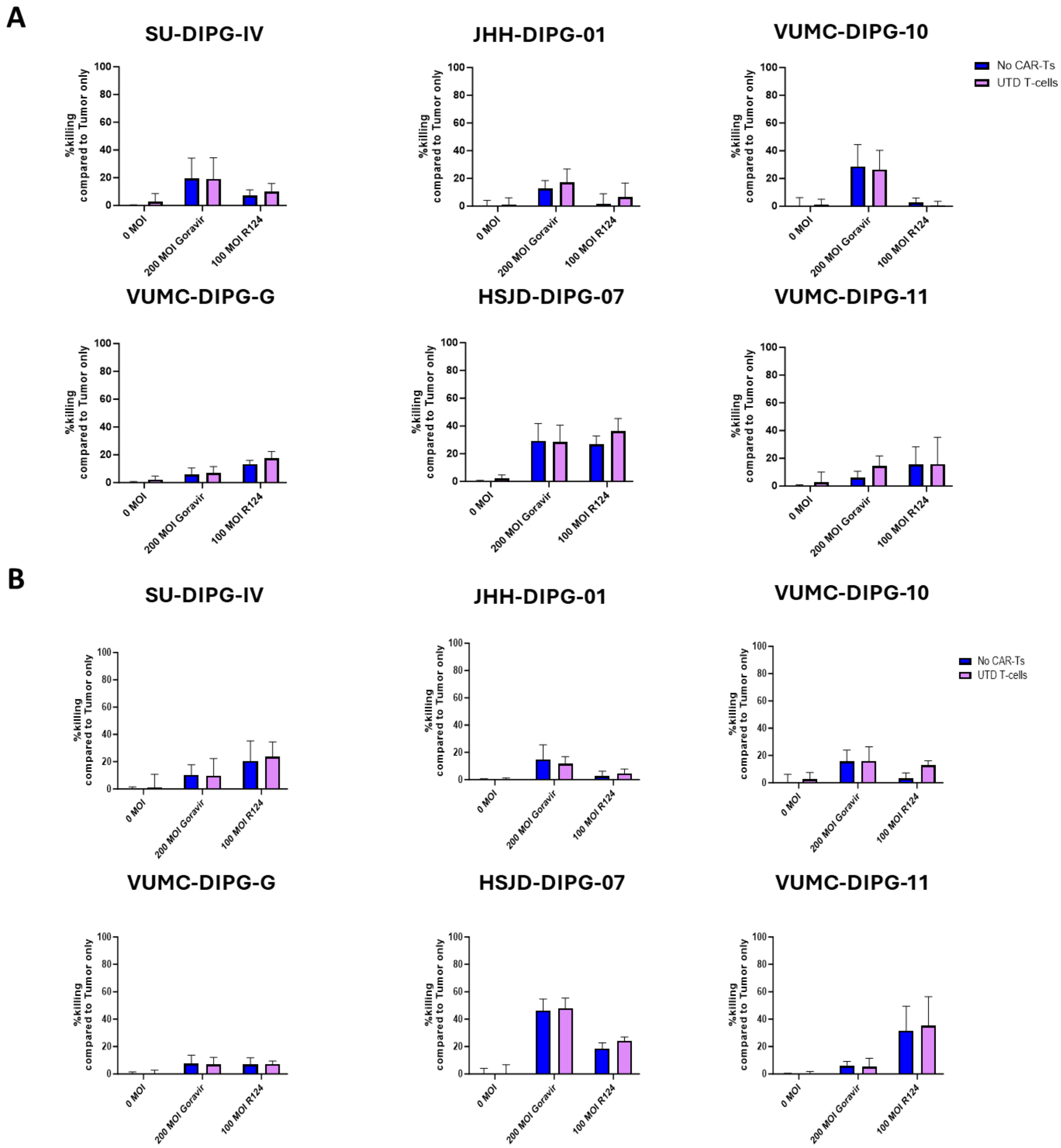


Figure S4. OV pre-infection of DMGs does not enhance *in vitro* anti-tumor killing by untransduced T-cells.

A) Anti-tumor activity of either Goravir or R124 or untransduced T-cells (UTD) monotherapies or combination of UTD T-cells with Goravir or R124 as assessed by Flow cytometry in *in vitro* co-culture assays. **B)** Anti-tumor activity of either Goravir or R124 or UTD monotherapies or combination of UTD

T-cells with Goravir or R124 as assessed by Flow cytometry in *in vitro* co-culture assays. For **B** and **C** each bar and error bar represents the mean ± SD of n=6 values. Statistics were assessed with two-way ANOVA and corrected for multiple comparisons by Tukey test. * , $p<0.05$; **, $p<0.01$; ***, $p<0.001$; ****, $p<0.0001$.

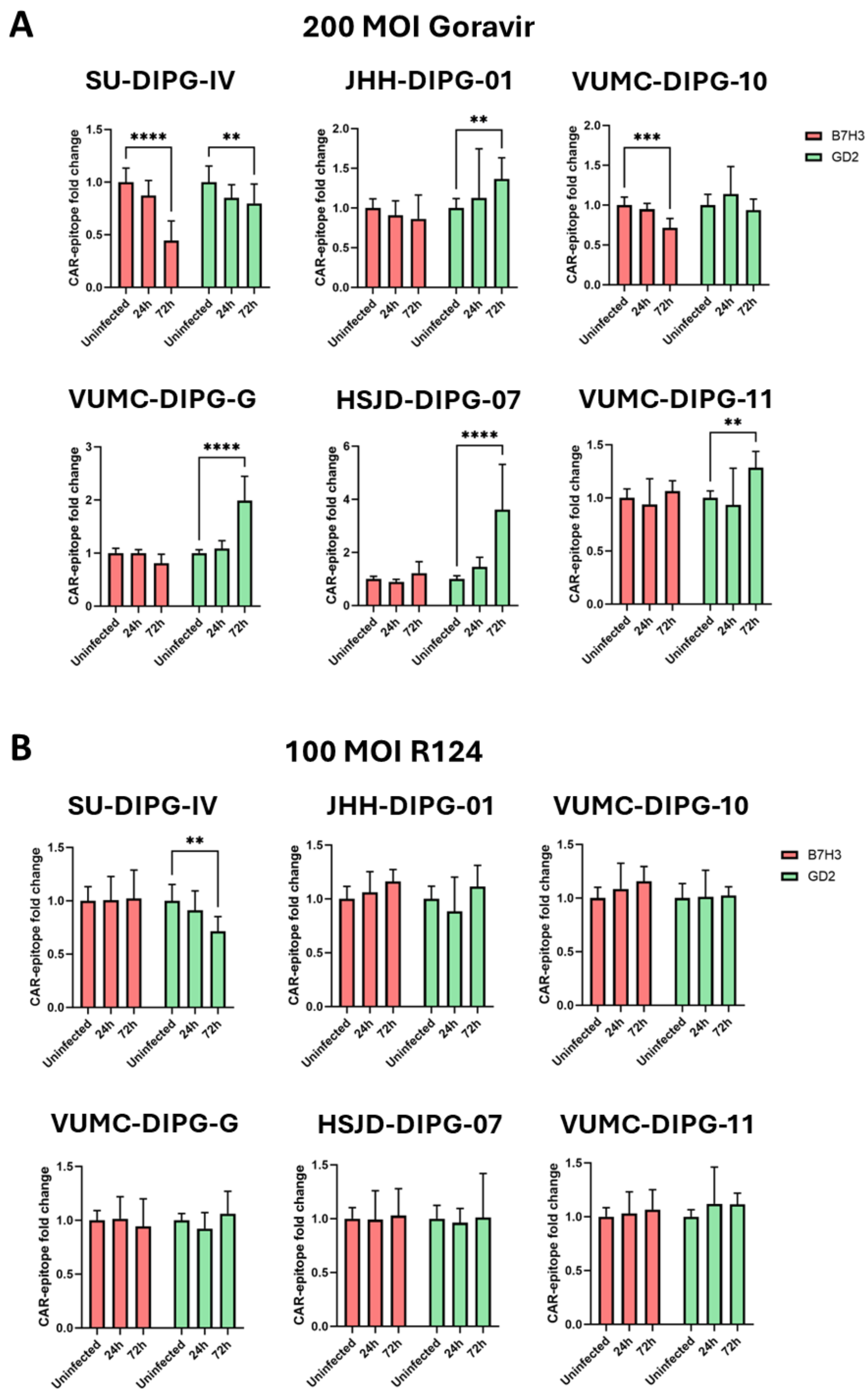


Figure S5. B7H3 and GD2 expression mainly change across Goravir-infected DMG cultures over time.

Fold change expression of B7H3 (red) and GD2 (green) on DMG cultures after 24 and 72 hours infection with **(A)** 200 multiplicity of infection (MOI) Goravir or **(B)** 100 MOI R124. Data were normalized on B7H3 or GD2 expression per uninfected DMG controls after 24 and 72 hours of incubation. Each bar represents the mean ± SD of n=7 values for 24 hours and n=8 for 72 hours. Statistics were assessed by one-way ANOVA and Tukey correction. * , $p<0.05$; **, $p<0.01$; ***, $p<0.001$; ****, $p<0.0001$.

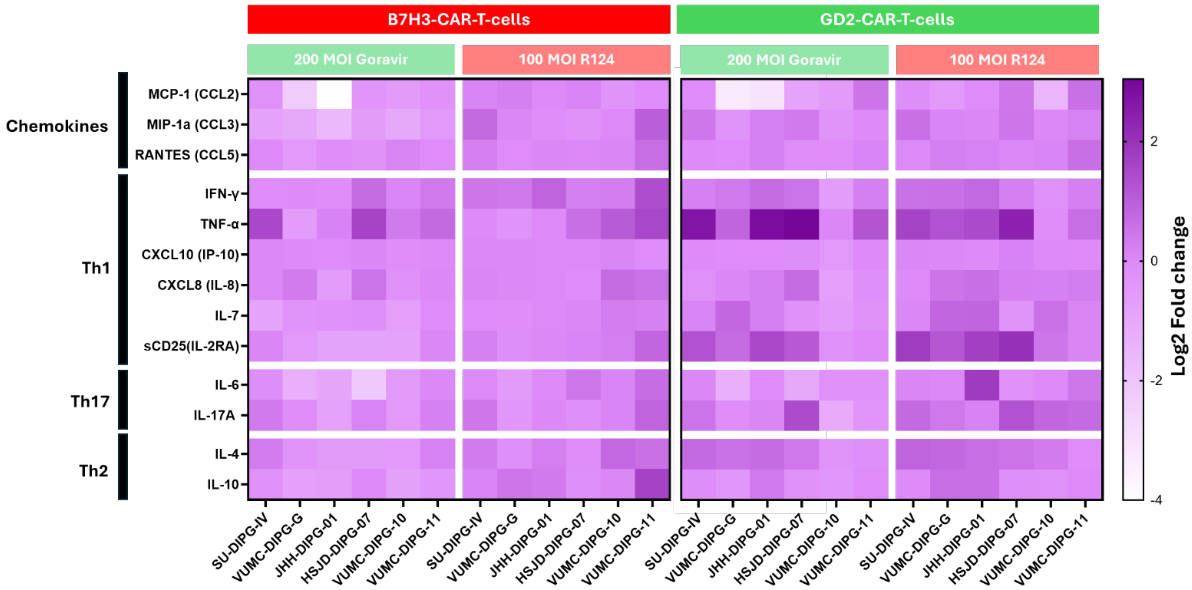
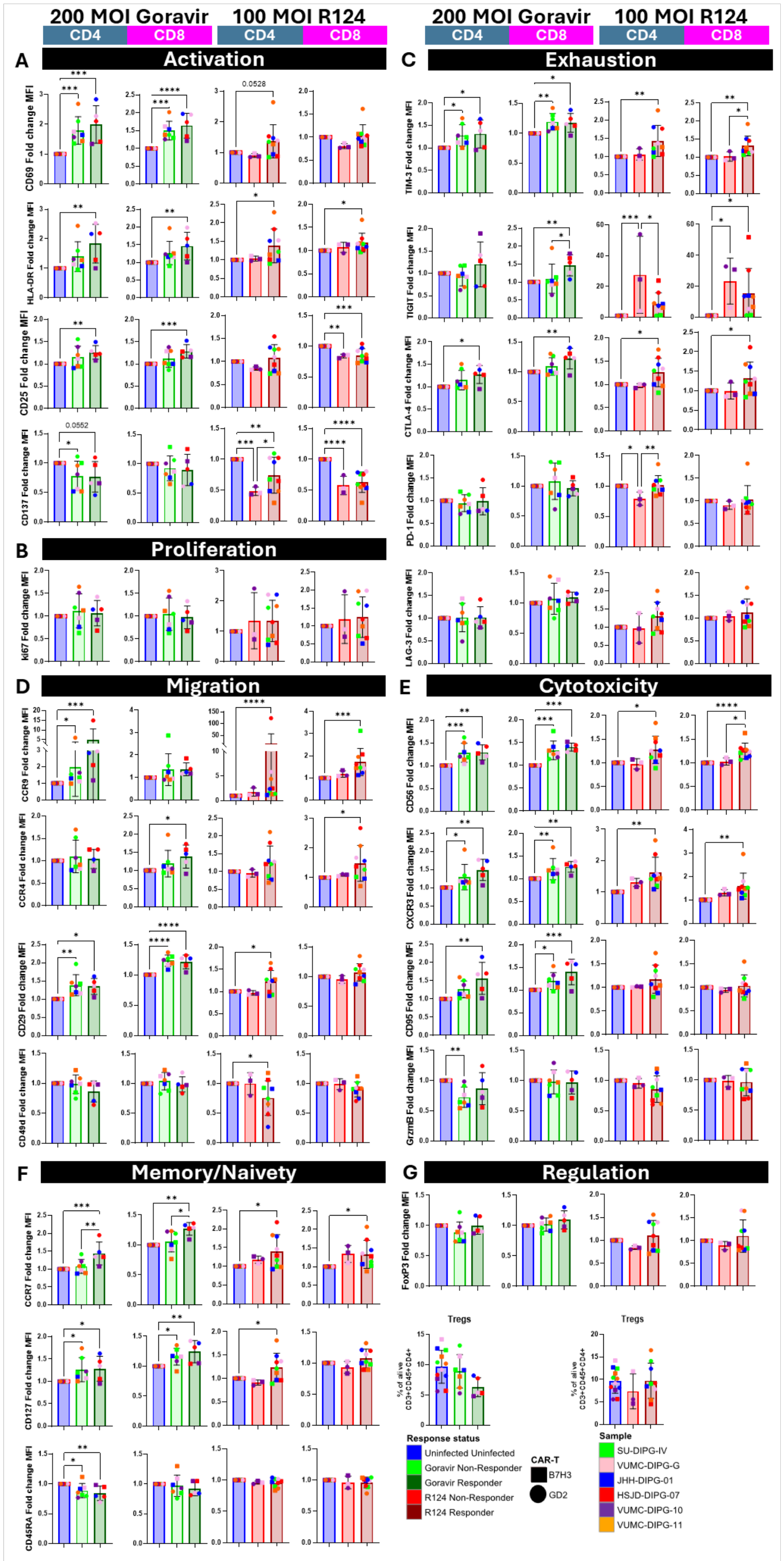


Figure S6. CAR T-cells provide a highly heterogeneous milieu of chemokine and cytokine changes per DMG co-culture pre-infected with Goravir or R124.

Chemokine and cytokine changes detected by Legendplex^{TM} and IFN-γ ELISA from co-culture supernatants of Figure 3B & C. Data are presented as the mean Log2 (Fold-change) per chemokine and cytokine measured from B7H3 (left) and GD2 (Right)-co-cultures pre-infected with Goravir (light green) or pre-infected with R124 (light red). Each dot represents a value from n=2 (n=3 for IFN-γ) independent experiments with the chemokine/cytokine measured from co-cultures of DMG uninfected cultures as baseline.



200 MOI Goravir
100 MOI R124
CD4
CD8
A
Activation
B
Proliferation
C
Exhaustion
D
Migration
E
Cytotoxicity
F
Memory/Naivety
G
Regulation
Tregs
Response status
Uninfected Uninfected
Goravir Non-Responder
Goravir Responder
R124 Non-Responder
R124 Responder
CAR-T
B7H3
GD2
Sample
SU-DIPG-IV
VUMC-DIPG-G
JHH-DIPG-01
HSJD-DIPG-07
VUMC-DIPG-10
VUMC-DIPG-11

Figure S7. OV-infected DMGs induce B7H3 and GD2 CAR T-cell specific phenotypic changes across CD4 and CD8 subsets, increasing CAR T-cell functionality.

Mean fluorescent intensity (MFI) fold change comparisons across various aspects of B7H3 and GD2 CAR T-cells among Goravir or R124 Responders, Non-Responders and uninfected DMG (SU-DIPGIV, VUMC-DIPG-G, JHH-DIPG-01, HSJD-DIPG-07, VUMC-DIPG-10 and VUMC-DIPG-11) co-cultures : **A**) activation markers: CD69, HLA-DR, CD25 and CD137, **B**) proliferation marker ki67, **C**) exhaustion markers: TIM-3, TIGIT, CTLA-4, PD-1 and LAG-3, **D**) migration markers CCR9, CCR4, CD29 and CD49d, **E**) MFI cytotoxicity markers: CD56, CXCR3, CD95 and GrzmB, **F**) memory markers: CCR7 and CD127 and naivety marker CD45RA, **G)** regulation marker FoxP3 (top) and population percentages comparisons of the T-cells Tregs (CD127low, CD25high and FoxP3high) from the total population of alive CD3+CD45+CD4+ cells. Each box and bar represents the mean ± SD of MFIs from CAR T-cells co-cultures, with each point color-coded to represent the fold change MFI for each DMG co-cultured with B7H3 (square) (n=18) or GD2 (circle) (n=18) CAR T-cells across uninfected, Non-Responders and Responders of either Goravir or R124 pre-infection using the expression of the respective CAR T-cells co-cultured with each uninfected DMG as baseline. Statistics were measured with repeated measures one-way ANOVA corrected with Tukey test for parametric data, while for non-parametric data Friedman paired test corrected with Dunn's test for multiple corrections was used. * , $p<0.05$; **, $p<0.01$; ***, $p<0.001$; ****, $p<0.0001$.